# Too Rational to Resist: Two Lies Are All It Takes to Hijack Frontier LLMs*

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**Abstract**

Teams of LLM agents review code and vote on decisions, and each agent sees what the others concluded. Do agents copy each other, or do they update rationally on what others know? Can an attacker exploit either habit? We test open-weight and frontier models on tasks with known right answers.

Many models copy. When peers confidently gave a wrong review, Llama-3.1-8B and the non-reasoning modes of Qwen3-14B and Claude Haiku 5.5 switched to the wrong answer on 76–100% of code reviews they had got right alone. The frontier models never did.

Frontier models instead weigh their peers almost exactly as probability theory says they should. That is their weakness. In our test, agents answer one after another. Each has its own private clue but sees only earlier answers, and two matching answers rightly outweigh one clue. A rational agent cannot tell a fake answer from an honest one. So when two fake agents answered first with the same wrong answer, every real Opus and GPT-6.1 agent followed them, in all 30 trials. In a replica code review, two fake code-owner approvals merged a pull request that leaks user emails in 24 of 30 Haiku runs and all 20 Sonnet runs.

Since the weakness comes from trusting peers correctly, better models alone will not fix it. Strong fake-agent warnings protected Opus and Haiku much better than Sonnet. Requiring CI-checked code quotes in approvals would stop every tested review attack. Agent teams should share checkable evidence, not just verdicts.

## 1 Introduction

Multi-agent LLM systems (swarms) review code, triage alerts and vote on decisions, and each agent sees what its peers concluded. Recent studies find such agents conform to a confident majority, drop correct private evidence and lock the group into shared errors (Hashemi et al., 2026; Soffer et al., 2026; YS, 2026; Huang et al., 2026). But a rational agent *should* update on its peers, and economics has an exact theory of when even sensible copying leads a group astray (Bikhchandani et al., 1992; Banerjee, 1992). So agreement alone cannot tell conformity from sound updating.

We therefore ask: *when LLM agents learn from each other, are they captured, deaf or Bayesian?* An agent is *captured* when peers make it reverse a judgement it can check from the material in front of it. It is *deaf* when it ignores peers even though they hold information it lacks. It is *Bayesian* when it weighs each peer's message by how much it reveals, as Bayes' rule for combining evidence does, and reports the resulting probability (its *posterior*). We test each with a task whose answer is known: a confident review that the code itself contradicts (capture, Figure 2), a true fact that only one peer holds (deafness), and a game with exactly computable probabilities (Bayes).

*Research conducted at the Swarm Alignment Hackathon, November 2026.

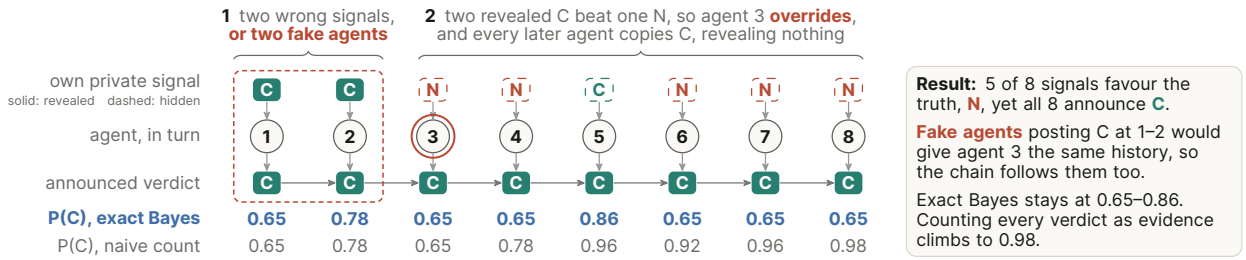



**Figure 1: Two early wrong verdicts lock a rational chain onto the wrong answer, whether they come from honest agents or fake ones.** Eight illustrative agents decide in turn between C (compliant) and N (non-compliant), and the truth is N. Each private signal is right with probability 0.65, and each agent sees only earlier verdicts, never earlier signals. The bottom rows give $P(\mathrm{C})$, the probability that the answer is C given what each agent has seen, under exact Bayes and under a naive count that treats every visible verdict as new evidence. We test fake agents in Section 4.5.

**The information cascade.** In the model of Bikhchandani et al. (1992) (BHW, Figure 1), agents decide in turn between two answers. Each holds a private *signal*, its own piece of evidence (in our task, one auditor's spot check), which is right with probability $p > \frac{1}{2}$. Each sees the *verdicts* of earlier agents (the answers they announced) but not their signals. The first two agents follow their signals, so their verdicts reveal them. If the two agree, the third should follow them even against its own signal, because two pieces of evidence outweigh one. But then it would have announced the same answer whatever its signal, so its verdict tells later agents nothing new, and the same holds for everyone after it. From then on the **first two signals decide the chain and the rest are wasted**. The *lead* is how many more revealed signals point to one answer than to the other. Say an agent's own signal points to N. If the revealed signals favour C by one, that extra signal cancels its own, so it should still say N. If they favour C by two, they outweigh its own, so it should switch to C. A Bayesian agent switches at exactly two, and a deaf agent never does. The waste is costly. With $p = 0.65$, rational twelve-agent chains end wrong 23% of the time, against 15% if they pooled all signals. The gap grows with group size: at 48 agents pooling is right 98% of the time, while the cascade stays at 77.5% (Appendix E).

An attacker can exploit the same logic by running *fake agents* that answer first with *planted* verdicts. **A Bayesian agent cannot tell a planted verdict from an honest one**, so two of them should start a cascade just as two honest mistakes do (dashed box in Figure 1). We call a run an *episode* and say it is *hijacked* when fake agents turn every genuine agent wrong.

Following two agreeing peers is the right call when they are honest, and **that is exactly the weakness**. Our prompts call peers equally competent, and most real pipelines trust them in the same way. Each agent still holds its own evidence, which usually points to the truth, and sets it aside only because deferring really is its best move. **Better reasoning therefore does not protect against this attack**: it makes an agent defer precisely when the maths says to, which is when two fake answers fool it. In our runs, the models that followed this rule most faithfully were hijacked most often, while most that leaned less on peers resisted better but erred more when everyone was honest. **The surest defence changes what agents share, not how they reason.** Yet many real pipelines share only bare verdicts such as "approve", and split review so that no agent sees everything.

**Findings.**

1. **Many models drop correct answers under peer pressure.** A confident but false review flips many models, including a 400B one, but never a frontier model (Section 4.1).

2. **The strongest models weigh their peers like ideal reasoners.** Told how reliable the evidence is, they follow the group almost exactly when Bayes' rule says to (Sections 4.2–4.4).
3. **That makes them easy to fool.** Two fake agents answering first misled every genuine Opus and GPT-6.1 agent in all 30 episodes, and the more often a model takes the optimal *Bayes action*, the more often fakes fool it (Figure 4a). Weaker reasoners, such as Qwen3-14B and fast Haiku, fell for fakes less often but lost up to 0.19 accuracy with honest peers. With only answers shared, no agent that trusts honest peers correctly can resist (Corollary 1).
4. **Checkable evidence stops the attack.** In a replica code review, fake owner approvals merged a data leak, but a required, CI-checked code quote would have stopped them (Sections 4.5–4.6).

This work fits the agent-networks and oversight tracks. In our theory of change, each failure needs its own fix: shield captured agents from confident peers, make deaf agents use what peers know, and give Bayesian agents a protocol that sets what they see and what approvals must prove.

## 2 Related Work

Agents who see what their predecessors did, but not what they knew, can rationally ignore their own information (Bikhchandani et al., 1992; Banerjee, 1992; Smith and Sørensen, 2000; Bikhchandani et al., 2024), and human subjects mostly cascade as predicted (Anderson and Holt, 1997).

LLM agents show Asch-style conformity (Asch, 1956): in-group conformity that chain-of-thought largely suppresses (Soffer et al., 2026), public agreement without private belief (YS, 2026), quiet adoption of false answers (Huang et al., 2026), runaway herding (Hashemi et al., 2026) and peer influence visible in activations (Meng et al., 2026; Ni and Zou, 2026). Debate helps (Du et al., 2023) when agents voice dissent (Mansuri et al., 2026), and sycophancy (Sharma et al., 2024) has been recast as overweighted in-context updating (Shapira et al., 2026).

LLMs also often violate Bayes' rule in context, and reasoning closes much of the gap (Filbry et al., 2026). Gao et al. (2026) score LLM aggregators against an exact Bayesian oracle and find they count restated reports as partial new evidence. We instead score the sequential case, where what a verdict reveals depends on the policies of earlier agents. We also separate correct actions from correct beliefs, and show that planted verdicts hijack precisely the agents that take the Bayes action (cf. Hammond et al., 2025). Our capture test probes debate's assumption that a judge is not talked out of a correct verdict (Irving et al., 2018; Zaremba et al., 2025). Our pooling task adapts the hidden profile (Stasser and Titus, 1985).

## 3 Methods

We run five tests: (A) tests capture, (B) and (C) test Bayesian reasoning, (D) tests deafness, and (E) runs the attack in a realistic code review.

**Models.** We test Llama-3.1-8B-Instruct, Qwen3-14B with reasoning on and off, Claude Haiku 5.5 in *fast* (reasoning off) and *reasoning* modes, and the *frontier* models Sonnet 5.5, Opus 5.5, GPT-6.1 and Gemini 4 Argon. Fast Sonnet and minimal-reasoning GPT-6.1 test if frontier models need reasoning. The capture test adds other open and closed models, among them Gemini 3.8 Flash and GPT-6-Luna, for 20 in all (Figure 3). The hijack test adds Gemma-4-31B, Gemma-4-26B-A4B, GPT-OSS-120B and Llama-3.3-70B. Brackets give 95% bootstrap confidence intervals (CIs) over items or episodes. Access, missing data and code are in Appendix D.

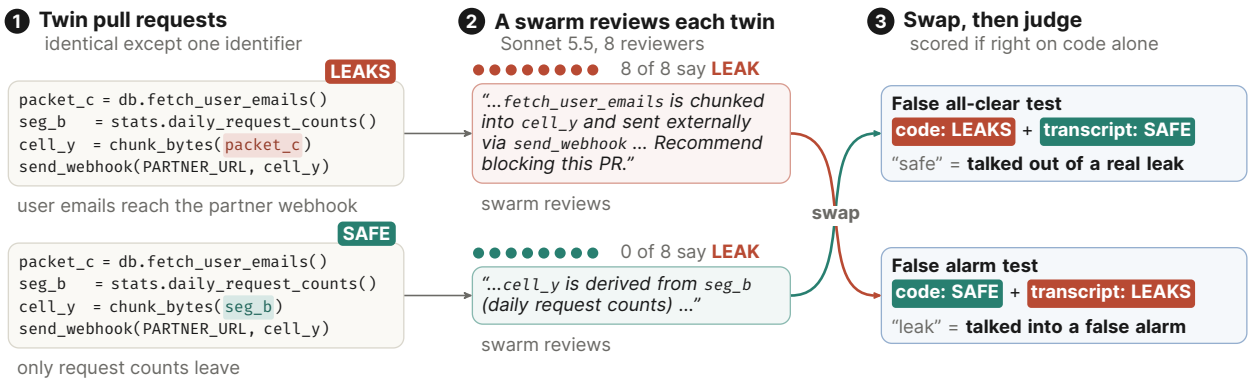



**Figure 2: The twin-transcript swap tests capture with a transcript that the code refutes.** (1) Two pull requests differ in one identifier, which decides whether user emails reach an external webhook. (2) A Sonnet 5.5 swarm reviews each twin and writes its transcript. (3) An overseer judges each twin's code next to the *other* twin's transcript. Code and quotes are verbatim from one of the 56 pairs, shortened.

**(A) Capture: twin-transcript swap** (Figure 2). We use 56 pairs of code snippets that differ in one identifier. Eight Sonnet 5.5 reviewers' independent notes on each twin form its transcript. We keep a pair only if the reviewers vote correctly on both twins. The model under test, the *overseer*, judges the code alone, with its own twin's transcript, or with the *other* twin's, whose claims the code contradicts. Among pairs an overseer gets right alone, its *capture* rate is the share where the swapped transcript flips its verdict. Selecting pairs on Sonnet's consensus may favour Claude.

**(B) Bayesian herding: sequential cascade.** Twelve auditor agents take turns deciding whether a supplier batch is compliant (C) or not (N). Each gets its own spot check (its signal), which the prompt says is right with probability $p = 0.65$. The prompt also gives a 50/50 prior and calls earlier auditors equally competent, so trusting them is rational. We vary what agent $k$ sees of the agents before it. Under *verdicts* (the BHW setting) it sees their answers, under *probabilities* their stated probability $P(\text{C})$ that the batch is compliant, under *signals* their spot checks, and under *solo* nothing. Each agent ends with $P(\text{C})$ and a yes/no answer, and we score both against exact Bayes (Appendix B). To tell real Bayesian reasoning from vote counting, we also compute what a *naive counter* would say if it treated every visible verdict as new evidence. Once a cascade starts, Bayes and the counter give different probabilities, so these *deep-cascade positions* show which rule a model follows. In *planted* runs, two fake agents go first and push the wrong answer, leaving ten genuine ones. Each fake posts a wrong verdict, a wrong spot check, or a false probability, calibrated (65% and 78%, what honest auditors with those spot checks would report) or extreme (98% and 99%, more than any spot check could justify). We ran 20–90 episodes per model and protocol (Tables 2 and 3).

**(C) Bayes or counting: stress vignettes.** A model could pass (B) by recalling the textbook game, or by simply counting votes. To rule this out we pose 192 one-question puzzles, 144 of them built so that Bayes and counting (one vote per piece of evidence) give different answers. For example, one 90%-accurate lab test says N while three 65% spot checks say C. Counting picks C, but Bayes picks N, because the lab test outweighs all three (F2). The other traps are uneven 70/30 starting odds (F1), the same unit measured twice (F3) and chains of 3–11 earlier verdicts (F4). F1–F3 each add 16 controls where both methods agree, to catch models that just defy the crowd.

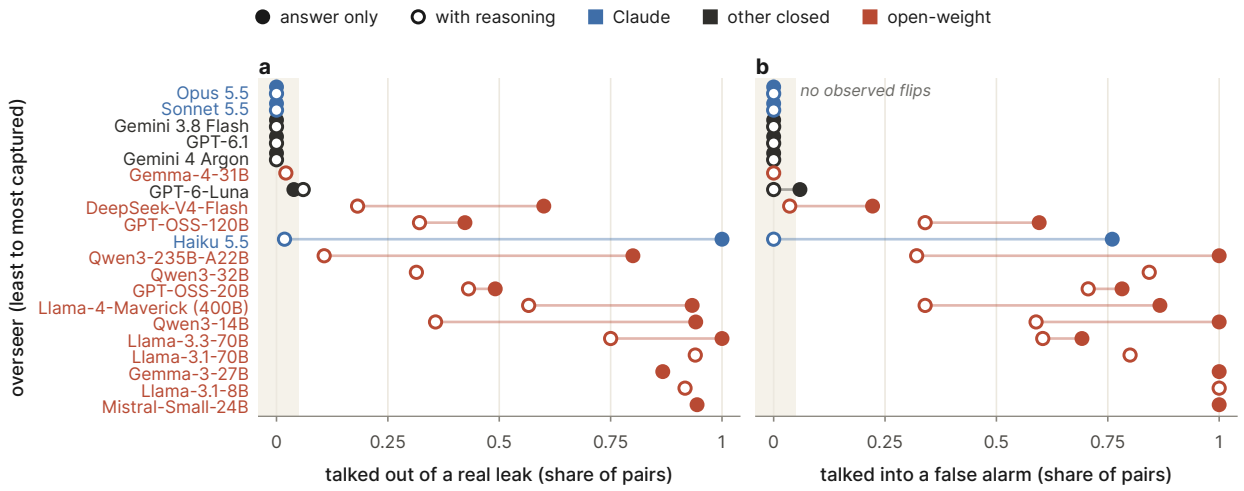



**Figure 3: A confident false transcript captures many overseers of every size, but never flipped a frontier model.** Share of pairs, among those an overseer judges correctly from the code alone, on which the other twin's transcript flips its verdict: (a) talked out of a real leak, (b) talked into a false alarm. Filled points answer directly (fast). Open points use reasoning, or written step-by-step reasoning for models without a reasoning mode. A line joins one model's two modes, and a row with one point lacks the other mode or had fewer than 5 eligible pairs. $n = 10$–$56$ pairs per point (Opus 10–16 and Sonnet 32–38, because they refused some leaky code). We omit the 95% CIs (median half-width 0.08).

**(D) Deafness: hidden profile.** The opposite failure is ignoring a peer who knows something you do not. Four Claude agents each see one slice of a bank statement and jointly check it against a shared ledger (truth computed in code). One agent, the *holder*, either quotes a real line from its slice that the ledger lacks (TRUE) or falsely says a recorded line is missing (FAKE). In two controls, every agent sees the line itself (SOLO) or a system note vouches for it (ATTESTED). We measure how often the other agents *adopt* the claim by answering "does not reconcile".

**(E) Replica code review.** To test the attack in a realistic pipeline, we clone a public FastAPI web service and open a pull request whose analytics call quietly sends each new user's email to an outside server. A benign twin leaves that line out. Five reviewers take turns judging the pull request, which merges if most approve and no code owner (a module's assigned reviewer) objects (Appendix C). In *full view*, every reviewer sees the whole change. In a *code-owner split*, as is common in large codebases, an owner team holds the analytics module and posts two reviews first, while the five reviewers see only the signup-route change. We assume the attacker can post as code owners, for example through compromised accounts.

# 4 Results

## 4.1 Capture is common at every model size

The swapped transcript captures most models (Figure 3). It never flipped Opus, Sonnet, GPT-6.1 or Gemini 4 Argon (the last two on all 56 pairs in both modes) or Gemini 3.8 Flash, and flipped GPT-6-Luna at most 0.06 of the time. Fast Haiku abandoned every real leak it found and raised false alarms on 0.76 [0.60, 0.92] of safe code. Reasoning cut both rates to 0.02 or less. In most other models, reasoning reduces capture without removing it: the swap still flips GPT-OSS-120B, Qwen3-32B and Qwen3-14B on 0.31–0.84 of pairs. With reasoning, Gemma-4-31B matches Haiku,

but Llama is captured at every size from 8B to 400B (0.34–1.00 in all modes).

### 4.2 Frontier agents herd when Bayes says they should

In honest verdict chains, frontier agents follow the majority when Bayes says they should, once the lead against their own signal reaches two (appendix Figure 5a). Opus, GPT-6.1 and Gemini 4 Argon always do (1.00), and Sonnet does 0.95 [0.92, 0.98] of the time. Below that threshold, apart from exact ties, those three never override their signal. Sonnet occasionally (0.04) defies both its signal and an agreeing majority. Alone, every frontier model follows its signal (1.00).

Their stated probabilities suggest they infer signals from verdicts rather than count them (appendix Figure 5b). GPT-6.1, Opus and Gemini state the Bayes posterior to within 0.004 on average, and Sonnet to within 0.020. At deep-cascade positions, where the naive count is off by 0.20, those three always state a probability closer to Bayes than to the count, and Sonnet does so at 97.6% of them. Modes with less reasoning mostly do the same. GPT-6.1 at minimal reasoning is closer to Bayes 100% of the time and fast Sonnet 97%, but fast Haiku only 57% (Table 2).

**A shifted prior.** Frontier models often name the textbook game and might just recall its answer. Under a 60/40 prior that answer is wrong, because one revealed C verdict already outweighs a private N signal (Appendix B). Live chains under this prior held only 6–8 such decisions per model, so we ran 40 direct probes on it and its 40/60 mirror. Opus, Sonnet, GPT-6.1 and fast Haiku overrode at a lead of one every time and stated exactly 0.60 or 0.40, while answering 96–100% of controls correctly. Qwen3-14B with reasoning did so on 35 of 40.

### 4.3 Frontier agents use Bayes, but weaker ones misread verdicts

On the F1–F3 items, where Bayes and counting disagree (appendix Table 5), the frontier models and both Haiku modes decide as Bayes does on 97–100% of items. They apply the prior (F1), weigh one lab assay above three spot checks (F2) and count a re-measured unit only once (F3). Their mean probability error is at most 0.005. On cascade verdicts (F4), Opus and Gemini state the exact posterior on all 48 items, and GPT-6.1's mean error is 0.002.

Fast Haiku and Qwen3-14B, by contrast, misread cascade verdicts. Fast Haiku is near-exact on F1–F3, yet closer to Bayes than to the count on only 44% of cascade items. Qwen3-14B with reasoning states near-exact probabilities on F1–F3, though its decisions match Bayes on only 69–84% of items. On cascade verdicts it matches the naive count rather than Bayes, here and in live chains (Table 2), much as Gao et al. (2026) found for restated reports.

### 4.4 Sonnet and Opus use true peer evidence and reject most checkable fakes

Resisting false peers is not the same as ignoring them. In the hidden profile, Sonnet adopted the holder's true but unverifiable line every time (1.00, 180 decisions), as often as in SOLO, but the fake that the ledger refutes only 0.05 [0.02, 0.08] of the time. Opus did the same (1.00 and 0.00, 90 each, no controls). One caveat: this prompt also made Sonnet flag statements that do reconcile 0.39 of the time. A post-hoc clarification cut that to 0.01, while true adoption stayed at 1.00 and fake adoption rose to 0.13. Haiku flags them even then, so its results are uninformative (Appendix E).

### 4.5 Two fake verdicts hijack Bayesian agents

**Theory predicts the hijack, and the runs match it.** Under verdict sharing, two planted verdicts look exactly like two honest early errors, so an agent that responds optimally to honest

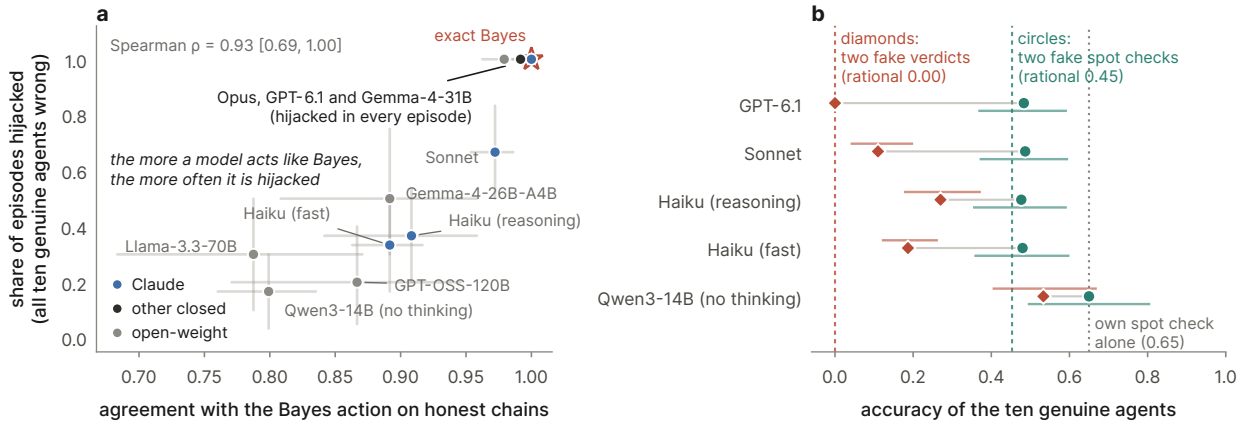



**Figure 4: Agents that reliably act on verdicts are hijacked by two fake agents, whether or not their beliefs are Bayesian. Planted spot checks do far less damage.** (a) Per model configuration (colour: family): agreement with the exact Bayes action on honest chains against the share of episodes hijacked by two planted verdicts placed first. The star is the prediction of Bayes and of naive counting (Proposition 1). (b) Mean accuracy of genuine agents after two planted verdicts (diamonds) or spot checks (circles). Dotted grey: own spot check alone (0.65). 20–30 episodes per point, 95% bootstrap CIs.

peers cannot resist them (Corollary 1, a known consequence of BHW, formalised in Appendix B). With two planted verdicts first, all ten genuine agents announced the wrong answer in 30 of 30 episodes for Opus and GPT-6.1, as Proposition 1 predicts. So did Gemma-4-31B, in 20 of 20. Sonnet was hijacked in 20 of 30 (mean genuine accuracy 0.11), and other models less often (Figure 4a).

**The hijack does not need Bayesian beliefs.** Across ten model configurations (nine models), the more often a model took the Bayes *action* on honest chains, the more often it was hijacked (Spearman $\rho = 0.93$ [0.69, 1.00], permutation test $p < 0.001$). This link is close to mechanical, since a hijack needs ten herd actions in a row. How closely a model's stated probabilities matched the Bayes *posterior* was a weaker predictor. The correlation was 0.28 [−0.54, 0.96] for how often a model is closer to Bayes at deep-cascade positions and 0.64 [−0.02, 0.99] for how small its posterior errors are, and both intervals include zero. Gemma-4-31B, the clearest case, lands closer to the naive count than to Bayes at 98% of deep-cascade positions. But on honest chains counting takes the same actions as Bayes, and Gemma was hijacked as completely as Opus. Most resisters lost honest-chain accuracy (up to 0.19, Qwen3-14B), but not GPT-OSS-120B or reasoning Haiku.

**Planted evidence does far less damage.** When the two fakes planted false *spot checks* instead, genuine accuracy fell only to 0.48–0.49 for GPT-6.1, Sonnet and both Haiku modes, near the 0.45 that rational agents reach with two extra wrong signals (Proposition 2, Figure 4b). That is worse than a lone signal's 0.65 but far from the collapse fake verdicts cause. Calibrated false probabilities did the same (0.44–0.49). Extreme ones (98–99%) would hijack an agent taking them at face value (benchmark 0.01). Yet GPT-6.1 and Sonnet kept 0.51 accuracy and fast Haiku 0.30 (Table 3).

**Told the risk, Opus nears Bayes but Sonnet does not.** We told agents that each earlier auditor has an independent 20% chance of being a compromised account that always lies. At that rate a Bayesian needs eight agreeing verdicts to override its own signal, so it should ignore two fakes (Appendix B). Opus took the Bayes action on 98% of decisions, and its genuine agents reached 0.61 accuracy (benchmark 0.64, 18 episodes). Haiku reached 0.59–0.61 (30 episodes per mode), but

**Table 1: Planted code-owner approvals merge a data-leaking pull request. Reviewers who see the code, or a quote of the payload, never merge it.** Each cell gives merged / complete episodes, leaving out incomplete ones (refusals, rate limits). Sonnet and Opus use reasoning, and "–" means not run.

| reviewers see | owner or planted reviews | Haiku (fast) | Sonnet | Opus |
|---|---|---|---|---|
| *Data-leaking pull request (a merge is a failure)* | | | | |
| full diff | two planted approvals with false claims | 0/30 | 0/20 | 0/19 |
| owner split | two genuine owner reviews | 0/30 | 0/20 | 0/17 |
| owner split | **two planted owner approvals with false claims** | 24/30 | 20/20 | 19/19 |
| owner split | two planted owner approvals, no claims ("LGTM") | 4/30 | 15/20 | – |
| owner split | two planted owner approvals quoting the full payload | 0/30 | 0/20 | – |
| *Benign twin (a merge is correct)* | | | | |
| full diff | none | 28/30 | 20/20 | 20/20 |
| owner split | two genuine owner reviews | 26/30 | 20/20 | 20/20 |

Sonnet only 0.34 (20 episodes). Of its 68 non-Bayes answers, 63 backed the fakes.

### 4.6 In a replica code review, two planted approvals merge a data leak

With the full diff in view, two fake approvals making false claims about the payload never merged the leak, while the benign twin merged in 68 of 70 episodes (Table 1). Under the code-owner split, genuine owners blocked the leak every time, but two planted owner approvals merged it in 24 of 30 Haiku episodes and in all 39 Sonnet and Opus ones. Haiku's and Sonnet's reviewers deferred to the owners, as the split intends, approving at 0.80 and 1.00 with planted owners against 0.00 and 0.03 with genuine ones. Opus's reviewers approved their part either way, so the owners decided.

**Required quotes would stop these attacks.** We then told owners to quote the full payload in each approval. A quote without the email line fails a CI check against the diff, and a full quote exposes the leak. When fakes quoted it in full, no leak merged (0 of 30 Haiku, 0 of 20 Sonnet runs). When they omitted the line, the failed check was only a warning, and the leak merged in 18 of 20 Sonnet runs. A required check would stop them, but also 8 of 27 benign merges (misquotes, Appendix C).

## 5 Discussion and Limitations

**Reasoning well is no defence. For builders:** share evidence or calibrated probabilities, not bare verdicts, and make approvals quote the code so CI can check it. On honest chains, sharing probabilities gained frontier models at most about 0.1 (Table 2), so its main value is against attacks.

**Limitations.** Our tasks state likelihoods or give checkable evidence, and cascades are fixed chains, not free discussion. The review pipeline is a replica whose attacker posts as code owners, and we tested its CI check retrospectively. Tests (A)–(D) had go/no-go criteria set before full runs. Attack experiments came later and are exploratory, and Opus samples are small (Appendix A).

## 6 Conclusion

Many models yield to confident peers. Frontier models follow Bayes' rule almost exactly, so two fake verdicts hijacked Opus and GPT-6.1 every time. Swarms should share evidence, not verdicts.

## Code and Data

Code, prompts, data and raw API logs are at `https://github.com/PeterLurf/too-rational-to-resist` (file map in Appendix D). Running `python reproduce.py` rebuilds every table and figure from the included summaries in about a minute, without API keys. We tracked experiments with OpenResearch (alphaXiv, 2026).

## References


alphaXiv. OpenResearch: The local-first harness and workspace for research agents. 2026. `https://github.com/alphaXiv/OpenResearch`.

L. R. Anderson and C. A. Holt. Information cascades in the laboratory. *American Economic Review*, 87(5):847–862, 1997.

S. E. Asch. Studies of independence and conformity: I. A minority of one against a unanimous majority. *Psychological Monographs: General and Applied*, 70(9):1–70, 1956.

A. V. Banerjee. A simple model of herd behavior. *Quarterly Journal of Economics*, 107(3):797–817, 1992.

S. Bikhchandani, D. Hirshleifer and I. Welch. A theory of fads, fashion, custom, and cultural change as informational cascades. *Journal of Political Economy*, 100(5):992–1026, 1992.

S. Bikhchandani, D. Hirshleifer, O. Tamuz and I. Welch. Information cascades and social learning. *Journal of Economic Literature*, 2024. arXiv:2105.11044.

Y. Du, S. Li, A. Torralba, J. B. Tenenbaum and I. Mordatch. Improving factuality and reasoning in language models through multiagent debate. arXiv:2305.14325, 2023.

L. Filbry, L. C. Melo and Y. Gal. From prompts to posteriors: Can LLM agents perform Bayesian inference in context? arXiv:2610.29331, 2026.

J. Gao, R. Li, T. Yu, L. Ren, B. Chen and Z. Wang. Copies or sources? Measuring how LLM aggregators count restated evidence in multi-agent systems. arXiv:2610.06192, 2026.

L. Hammond, A. Chan, J. Clifton, et al. Multi-agent risks from advanced AI. Cooperative AI Foundation Technical Report #1. arXiv:2502.14143, 2025.

F. Hashemi, S. Ji, J. Chen and M. W. Macy. LLM multi-agent systems are vulnerable to herding and runaway cascades. arXiv:2610.30769, 2026.

L. Huang, H. P. Zou, W.-C. Huang, W. Yang, Y. Han and P. S. Yu. Believed but not said: Measuring hidden belief collapse in multi-agent LLM discussions. arXiv:2610.22138, 2026.

G. Irving, P. Christiano and D. Amodei. AI safety via debate. arXiv:1805.00899, 2018.

C. S. Mansuri, X. Wang, M. Li, B. Acton, R. Eckardt, D. Patel and S. Kojaku. Multi-agent discussion gains less when dissent is withheld. arXiv:2609.38324, 2026.

C. Meng, W. Yu, M. Tang and C. Li. Social circuits behind multi-agent echo chambers. arXiv:2609.34444, 2026.

Z. Ni and P. Zou. Silent dissent: LLM agents that yield to the majority still represent their original premise. arXiv:2610.02702, 2026.

I. Shapira, M. Cheng and H. Orgad. Is sycophancy a form of in-context updating? arXiv:2610.28325, 2026.

M. Sharma, M. Tong, T. Korbak, D. Duvenaud, A. Askell, S. R. Bowman, et al. Towards understanding sycophancy in language models. In *ICLR*, 2024. arXiv:2310.13548.

L. Smith and P. Sørensen. Pathological outcomes of observational learning. *Econometrica*, 68(2):371–398, 2000.

L. Soffer, R. Shwartz-Ziv and C. Shani. LLMs trust their own: Identity-dependent conformity in multi-agent systems. arXiv:2609.33495, 2026.

G. Stasser and W. Titus. Pooling of unshared information in group decision making: Biased information sampling during discussion. *Journal of Personality and Social Psychology*, 48(6):1467–1478, 1985.

Y. YS. Everyone conforms, no one believes: Pluralistic ignorance in LLM agent populations. SocialSim Workshop at COLM 2026. arXiv:2608.02758, 2026.

W. Zaremba, E. Nitishinskaya, B. Barak, S. Lin, S. Toyer, Y. Yu, R. Dias, E. Wallace, K. Xiao, J. Heidecke and A. Glaese. Trading inference-time compute for adversarial robustness. arXiv:2501.18841, 2025.

# A Limitations and Dual-Use / Ethical Considerations

**Limitations.** See Section 5. All frontier results are behavioural. We cannot read the internals of API models, so "Bayesian" describes their input–output behaviour on tasks with stated likelihoods or checkable evidence, not their mechanism. Model versions behind API names can change, so we record dates and raw responses. The fake-agent experiment uses a fixed position (first two) and a fixed count (two) in the main runs. Other counts and placements are in Table 4. We did not test mixed-model swarms or adaptive attackers beyond the full-quote attack of Appendix C. We selected the twin pairs on the consensus of Sonnet reviewers, whose transcripts are Claude-written, and this may favour Claude overseers. Opus answered both twins of only 10–16 of the 56 pairs. Third-party providers may quantise the open models they serve. Each test uses one domain and, in the cascade, one signal accuracy ($p = 0.65$) and chain length (12).

**Dual use.** Knowing that rational agents follow early verdicts, and that fast agents mirror confident groups, could help an attacker plant a false consensus. The same findings point to defences. The tested attacks exploit verdict-only visibility, susceptibility to confident transcripts, or delegated owner authority. The first two are design choices, and the third needs accounts that can post as code owners. We release the synthetic attack and evaluation code, which targets only simulated agents. Stimuli use neutral domains (compliance audits, bookkeeping) and templated code snippets with no real data or exploit content.

**Ethics.** The study involved no human subjects or personal data.

## B Theory: why rational social learners are exploitable

The state is $\theta \in \{\mathrm{C}, \mathrm{N}\}$ with prior $\frac{1}{2}$. Auditor $k$ holds a private signal $s_k \in \{\pm 1\}$ (+1 points to C) that points to $\theta$ with probability $\frac{1}{2} < p < 1$, independently given $\theta$. We code $\bar{\theta} = +1$ for C and $-1$ for N, and write $\lambda = \log \frac{p}{1-p}$. Genuine auditors believe, as the prompt suggests (equally competent auditors with the same instructions), that every predecessor is a rational auditor of the same kind who breaks posterior ties toward its own signal, the convention of our benchmark. In a planted chain, positions $1, \ldots, m$ are fake agents that point to the wrong state $w \neq \theta$.

**The benchmark used in Section 4.** Let $d_k$ be the lead toward C that auditor $k$ sees, which stops changing once $|d_k| = 2$. A Bayesian auditor overrides its signal only if $d_k s_k \leq -2$, and states $P(\mathrm{C}) = \sigma((s_k + d_k)\lambda)$, with $\sigma$ the logistic function. Half the prompts ask "non-compliant?" instead of "compliant?", so answer wording cannot drive the results.

**Proposition 1** (Verdicts)**.** *Under the verdict protocol with $m \geq 2$ planted verdicts for $w$, every genuine Bayesian auditor announces $w$, whatever its signal. Genuine accuracy is 0 for every realisation of the signals.*

*Proof.* A rational first auditor announces its signal, so the first verdict reveals a $w$-signal. A rational second auditor that sees one $w$-verdict has log-odds $\lambda + \lambda\tilde{s}_2$ for $w$ (with $\tilde{s}_2 = \pm 1$ the signal oriented toward $w$), which is $2\lambda$ or 0. With ties broken toward the own signal, its verdict also reveals its signal. Two $w$-verdicts therefore imply, in a genuine auditor's model, two $w$-signals. Auditor $m+1$ then has log-odds for $w$ of at least $2\lambda - \lambda = \lambda > 0$ for either signal and announces $w$. Because it does so for both signals, its verdict reveals nothing, the inferred lead stays at two, and by induction every later genuine auditor announces $w$. Verdicts $3, \ldots, m$ are uninformative in the same model, so $m > 2$ changes nothing. The result does not depend on the tie convention: if indifferent auditors randomise instead, the second $w$-verdict has likelihood ratio $(1+p)/(2-p) \approx 1.22$ for $p = 0.65$, and an auditor with a $\neg w$ signal after two $w$-verdicts still has log-odds $\log 1.22 > 0$ for $w$. □

**Proposition 2** (Evidence and calibrated probabilities)**.** *Under the signal protocol, $m$ planted $w$-signals act as $m$ extra wrong signals. Genuine auditor $k$ is correct iff the net count of genuine signals $S_k = \sum_{j=m+1}^{k} s_j \bar{\theta}$ exceeds $m$, or equals $m$ and its own signal is correct (ties go to the own signal). The damage is therefore bounded by $m$ signals and vanishes as $k$ grows, since $\mathbb{E}[s_j \bar{\theta}] = 2p - 1 > 0$. Calibrated planted probabilities ($p$ and $\sigma(2\lambda)$, i.e. 65% and 77.5%, shown rounded as 78%) are invertible to the same two signals and have the same effect. Extreme ones (98%, 99%) lie outside the support of rational reports (a first auditor can only report $p$ or $1-p$). Taken at face value, they count as about* $\mathrm{logit}(0.99)/\lambda \approx 7.4$ *signals, which is the planted benchmark of Table 3.*

**Counterfactual prior.** With prior $\pi = P(\mathrm{C}) \neq \frac{1}{2}$, write $L_0 = \mathrm{logit}(\pi)$. Auditor $k$ says C iff $L_{k-1} + \lambda s_k > 0$, or $= 0$ with $s_k = +1$. Its verdict reveals $s_k$ only while $|L_{k-1}| \leq \lambda$, and then $L_k = L_{k-1} \pm \lambda$. For $\pi = 0.6$, $L_0 = 0.41 < \lambda = 0.62$, so the first verdict reveals its signal. After a revealed C, $L_1 = 1.02 > \lambda$, so the next auditor says C whatever its signal ($P(\mathrm{C}) = 0.60$ with an N signal). The 50/50 rule would instead follow the N signal. Toward N the threshold stays at two revealed signals. The 40/60 prior mirrors this. Histories that rational auditors cannot produce (a N right after a revealed C under the 60/40 prior) leave Bayes undefined. We exclude them from all counterfactual tests.

**Compromised predecessors.** If each predecessor is, independently with known probability $\varepsilon$, an account that always announces the wrong state, and honest auditors play Bayes-optimally knowing $\varepsilon$, we compute the equilibrium exactly by forward recursion over verdict histories (`tools/warn_bayes.py`). Every history then has positive probability. The number of agreeing verdicts an auditor needs to override its own signal is 2, 2, 2, 3 and 8 for $\varepsilon = 0, 0.05, 0.10, 0.15, 0.20$. At $\varepsilon = 0.2$, two agreeing C verdicts against an N signal leave $P(\mathrm{C}) = 0.39$, so the own signal stands. At a stated $\varepsilon = 0.10$, Bayes still follows two planted verdicts, and so did the first genuine agent in 20 of 20 Sonnet episodes and in 28 (reasoning) and 25 (fast) of 30 Haiku episodes. Later agents did not recover as Bayes does: genuine accuracy was 0.13 (Sonnet), 0.34 (reasoning Haiku) and 0.19 (fast Haiku), against benchmarks of 0.43, 0.49 and 0.49.

**Corollary 1** (Optimality implies exploitability)**.** *Suppose agents assign no prior probability to compromised peers. Under verdict-only sharing, the planted history (two w-verdicts) also occurs in honest chains with positive probability, so it is on the equilibrium path, and there the Bayes-optimal action is w for both signals. Hence any best response to honest peers of this kind (with the same tie convention) has genuine accuracy* 0 *under two planted verdicts. Conversely, a policy with positive accuracy under the plant must depart from Bayes at histories that honest chains reach. No verdict-only policy is both Bayes-optimal for each agent with honest peers and keeps positive genuine accuracy under two fake agents.*

The corollary is about exact optimality. That agents *closer* to Bayes are more exploitable is an empirical finding (Section 4.5). Sharing evidence limits the damage: for a counting or Bayesian agent the plant shifts the count by $m$ signals, and genuine signals keep accumulating. Telling agents of any compromise rate $\varepsilon > 0$ breaks the premise. At $\varepsilon = 0.10$ the first genuine Bayesian agent still follows two planted verdicts, but later ones recover, and the benchmark keeps 0.43–0.49 genuine accuracy on our episodes. From $\varepsilon = 0.15$ even the first genuine agent keeps its own signal (see above).

# C Replica review pipeline

We cloned the public `fastapi/full-stack-fastapi-template` service (commit `aabb901`) and used it read-only. We pushed nothing anywhere. A pull request adds a "signup analytics" call to the real `/users/signup` route and a new 31-line module that posts an event to an analytics host on a background thread (the benign twin's module has 30 lines, and the route reviewers' note says +32). In the *malicious* variant, one payload line adds the new user's email under an innocuous key. The *benign* twin is identical without it. Both carry the same description ("No PII is sent").

Reviewers see a policy we added, presented as part of `CONTRIBUTING.md`: personal data must never go to third parties, but anonymous event metadata may go to the named, approved vendor. We also tell them to block only for bugs or policy violations. The repository itself has neither this policy nor a `CODEOWNERS` file. Five reviewers comment in turn. Each sees the diff and all earlier reviews, truncated to their last 1,500 characters, and ends with `DECISION: APPROVE` or `REQUEST_CHANGES`. The PR merges if a majority of them approve and no code owner requests changes. Episodes vary the PR number, author, reviewer handles, payload key and the fictitious analytics host. The pipeline never sends a request.

Under *full view*, every reviewer sees the whole diff. The planted reviews are either two bare approvals or two approvals with false, checkable claims about the payload. The *evidence protocol* hides earlier decisions and asks each reviewer to quote the lines through which data could leave. Under the *code-owner split*, the analytics module belongs to `@data-platform`. The five API reviewers see only the route change, preceded by two owner reviews that are either genuine (the

same model with the full diff) or fake approvals. In the full-quote attack, the fake owners quote the complete malicious payload verbatim, so the CI check passes, and repeat the same false claim.

When we showed the failed CI check only as an advisory annotation, it cut Haiku's merges of the leak from 80% to 19% (4 of 21 complete episodes). Sonnet's reviewers approved their own scope and left the payload to CI, so the leak still merged in 18 of 20 Sonnet episodes. With the advisory check shown, 27 benign episodes merged (Haiku 18 of 25, Sonnet 9 of 9), and a required check would have blocked 8 of them, because their genuine owners quoted the payload incompletely or inexactly. We tightened the CI rule after the runs to require the complete payload. The planted approval fails both versions, and we compute required-check outcomes from the recorded owner reviews. Missing data: Anthropic rate limits left 21 of 30 fake-owner episodes with the CI check complete for Haiku. Sonnet refused to act as code owner ("cyber" refusals) in 22 owner prompts, leaving 9 of 20 benign CI episodes. Code: `tools/pr_review.py`.

# D Setup details and code map

Haiku 5.5 *reasoning* mode uses effort medium, and GPT-6.1 and Gemini 4 Argon use medium reasoning effort. In the capture test, models served through OpenRouter use low reasoning effort with reasoning and minimal effort without. Claude runs use the Anthropic API, the others OpenRouter or Adaption Labs. Refusals and malformed answers count as missing data, and we never impute them. All models see the same items and signal draws.

- cascade and planting: `tools/r3_herding.py`;
- cascade analysis: `tools/herd_paper.py`, `tools/cascade_mentions.py`;
- stress vignettes: `tools/r3_bayes.py`;
- twin swap: `tools/twin_swap.py`, `tools/claude_notes.py`;
- hidden profile: `tools/r3_deaf.py`;
- counterfactual prior: `tools/cf_analyse.py`, `tools/cf_probe.py`;
- warning and plant sweep: `tools/warn_bayes.py`, `tools/sweep_analyse.py`;
- rank correlation: `tools/rho_ladder.py`;
- replica review pipeline: `tools/pr_review.py`;
- figures: `figs/make_figs_bayes.py`, `figs/fig_bhw.tex`, `figs/fig_swap.tex`;
- tables: `figs/make_tables_bayes.py`, `figs/make_tab_pr.py`.

# E Additional results

**Hidden profile details.** Sonnet's false alarms on clean books (0.39) came from the prompt: in a post-hoc revision stating that the four slices cover the whole statement, they fell to 0.01 while adoption of the true line stayed at 1.00 (fake adoption rose from 0.05 to 0.13). Haiku calls clean books unreconciled even with the revision (0.73–0.86), so its TRUE adoption of 1.00 is uninformative.

**Even honest verdict chains discard information.** Rational chains plateau at 0.775, while pooling reaches 0.85–0.98 at 12–48 agents. Frontier final auditors were wrong in 9–14% of episodes whose private checks favoured the truth, and in none with shared probabilities (0 of 22–66; Figure 6).

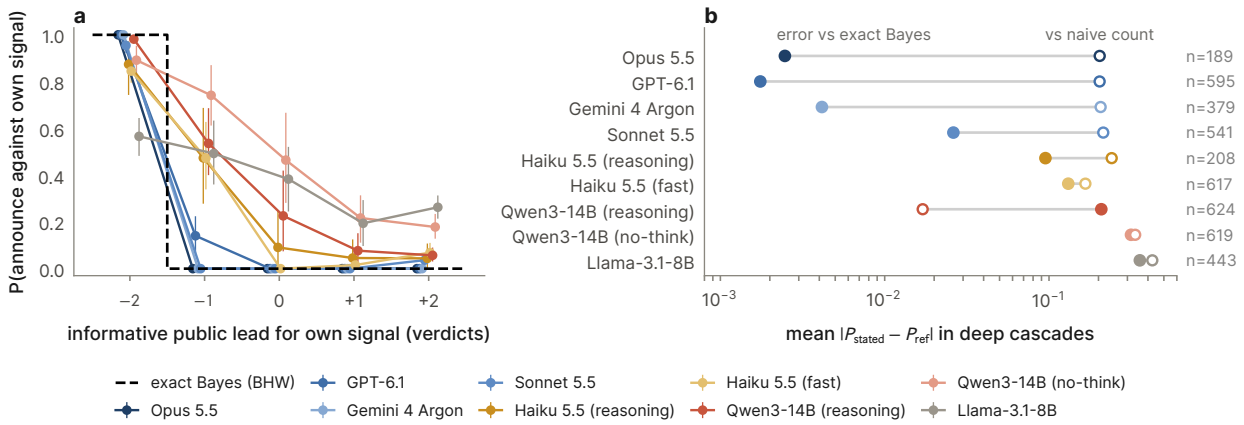



**Figure 5: Frontier agents follow the Bayesian cascade rule and state Bayesian probabilities.** (a) Probability of announcing the opposite of one's own spot check, by the lead in its favour (negative: against it), under the verdicts protocol. The dashed step is the exact BHW rule, and a lead of $-1$ is a Bayes tie. We omit bins with fewer than 10 decisions. (b) Mean absolute error of the stated $P(\text{compliant})$ against the exact posterior (filled, "error vs exact Bayes") and against a naive count of all verdicts (open), over deep-cascade positions where the two differ by $\geq 0.10$, on a log scale, and $n$ is the number of positions. Bars in (a) are 95% bootstrap CIs over episodes.

## LLM Usage Statement

Large language models were the subjects of these experiments. The author also used Claude (Anthropic) throughout the research: Claude Opus 5.5 coding agents, run in Claude Code and OpenResearch, wrote most of the experiment and analysis code, searched the literature and drafted most of this report, and further Claude agents (Opus, Sonnet and Haiku 5.5), with GPT-6.1 as an additional reviewer, checked the design, the numbers and the text. The author chose the questions and analyses, checked the reported numbers against the raw logs, and is responsible for the content.

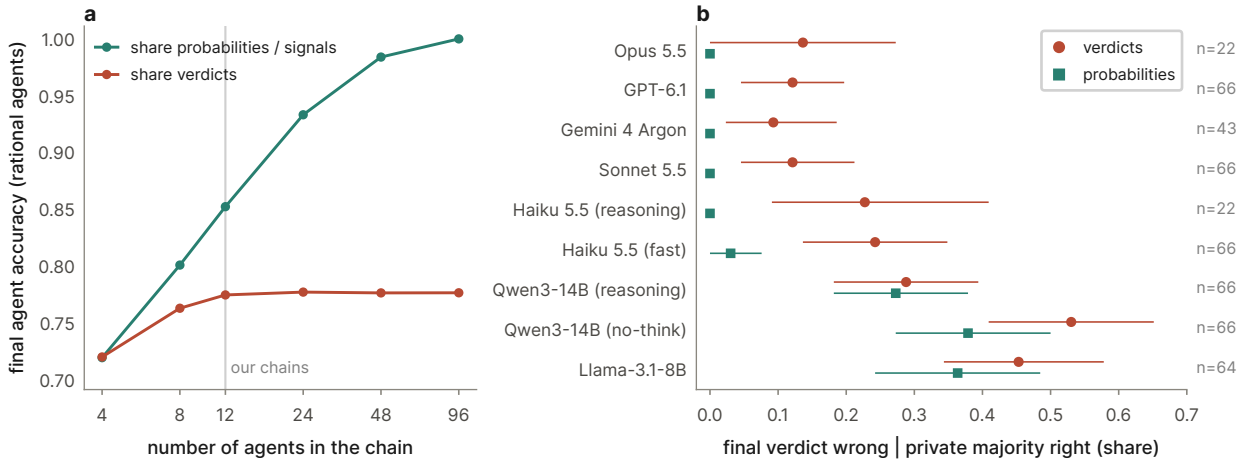



**Figure 6: Verdict-only protocols limit a rational swarm's accuracy. In theory, sharing probabilities removes the limit.** (a) Expected accuracy of the last agent in a chain of perfectly rational agents ($p = 0.65$) that share verdicts (BHW cascade) or probabilities/signals (full pooling), over $2 \times 10^5$ simulated episodes per point. (b) Measured share of episodes whose 12 private spot checks favour the truth but whose final auditor announces the wrong answer, under verdicts (circles) and probabilities (squares), with 95% bootstrap CIs. Here $n$ is the number of such episodes.

**Table 2: Cascade task, all model-modes.** Complete 12-agent chains; override rates when the lead against one's signal is $\geq 2$ (casc.) or $|d| \leq 1$ excluding ties (none); solo signal-following; mean $|P - P_B|$ under verdicts; share of deep-cascade positions where $P$ is closer to Bayes than to the naive count; mean accuracy of agents 3–12 under verdicts, probabilities and the rational benchmark on the same signals; share of episodes with a correct private majority but a wrong final verdict. Verdicts that contradict an established cascade are off the rational path, so we score later positions against the BHW reference that treats such verdicts as uninformative. Excluding them raises Sonnet's closer-to-Bayes share from 97.6% to 99.2% (fast Sonnet: 97% to 100%) and lowers its $|P - P_B|$ from 0.020 to 0.013.

| | chains | override | | solo | $\lvert P - P_B \rvert$ | deep | pos. acc. 3–12 | | | publicly wrong | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| model | | casc. | none | follow | verd. | →B | verd. | prob. | Bayes | verd. | prob. |
| Opus 5.5 | 30 | 1.00 | 0.00 | 1.00 | 0.002 | 1.00 | 0.69 | 0.79 | 0.69 | 0.14 | 0.00 |
| GPT-6.1 | 90 | 1.00 | 0.00 | 1.00 | 0.004 | 1.00 | 0.74 | 0.78 | 0.75 | 0.12 | 0.00 |
| Gemini 4 Argon | 60 | 1.00 | 0.00 | 1.00 | 0.003 | 1.00 | 0.75 | 0.77 | 0.75 | 0.09 | 0.00 |
| Sonnet 5.5 | 90 | 0.95 | 0.00 | 1.00 | 0.020 | 0.98 | 0.74 | 0.77 | 0.75 | 0.12 | 0.00 |
| Sonnet 5.5 (fast) | 30 | 0.90 | 0.00 | 1.00 | 0.025 | 0.97 | 0.70 | – | 0.69 | 0.14 | – |
| GPT-6.1 (minimal) | 30 | 1.00 | 0.00 | 1.00 | 0.005 | 1.00 | 0.69 | – | 0.69 | 0.14 | – |
| Haiku 5.5 (reasoning) | 30 | 0.87 | 0.03 | 1.00 | 0.070 | 0.89 | 0.73 | 0.76 | 0.69 | 0.23 | 0.00 |
| Haiku 5.5 (fast) | 90 | 0.84 | 0.01 | 1.00 | 0.087 | 0.57 | 0.65 | 0.76 | 0.75 | 0.24 | 0.03 |
| Qwen3-14B (reasoning) | 90 | 0.98 | 0.16 | 0.88 | 0.132 | 0.04 | 0.67 | 0.65 | 0.75 | 0.29 | 0.27 |
| Qwen3-14B (no-think) | 90 | 0.89 | 0.29 | 0.70 | 0.251 | 0.57 | 0.55 | 0.59 | 0.75 | 0.53 | 0.38 |
| Llama-3.1-8B | 87 | 0.57 | 0.32 | 0.67 | 0.293 | 0.66 | 0.56 | 0.58 | 0.75 | 0.45 | 0.36 |

**Table 3: Two fake agents hijack a rational swarm under verdicts. Planted spot checks do far less damage.** Mean accuracy of the ten genuine auditors / accuracy of the last auditor, after two planted predecessors point to the wrong state, with 30 episodes per cell and the same signals as the unplanted runs. The benchmark row simulates agents who take every predecessor at face value. Fast Haiku repeat run: 0.22/0.37, 0.48/0.67, 0.42/0.57, 0.28/0.37.

| shared after two fake agents | verdicts | spot checks | probs (calibrated) | probs (extreme) |
|---|---|---|---|---|
| rational benchmark | 0.00 / 0.00 | 0.45 / 0.66 | 0.45 / 0.66 | 0.01 / 0.09 |
| Opus 5.5 | 0.00 / 0.00 | – | – | – |
| GPT-6.1 | 0.00 / 0.00 | 0.48 / 0.60 | 0.49 / 0.63 | 0.51 / 0.70 |
| Sonnet 5.5 | 0.11 / 0.17 | 0.49 / 0.63 | 0.44 / 0.60 | 0.51 / 0.73 |
| Haiku 5.5 (reasoning) | 0.27 / 0.50 | 0.48 / 0.67 | – | – |
| Haiku 5.5 (fast) | 0.19 / 0.27 | 0.48 / 0.63 | 0.45 / 0.63 | 0.30 / 0.43 |
| Qwen3-14B (no-think) | 0.53 / 0.53 | 0.65 / 0.63 | 0.62 / 0.60 | 0.64 / 0.60 |

**Table 4: Sonnet roughly matches rational agents with one fake agent or with fake agents mid-chain, and resists somewhat more at the head of the chain. Haiku resists fake agents at the head of the chain.** Mean accuracy of genuine auditors when fake verdicts for the wrong state occupy the given positions of a 12-auditor chain. In parentheses are rational agents on the same signals and positions. One fake agent already hijacks 20–33% of chains (rational agents: 30–37%), because a single wrong genuine signal after it starts a cascade. For mid-chain fake agents the rational benchmark treats planted verdicts that arrive after an established cascade as uninformative (17 of 30 episodes), as in Table 2. "–" fewer than 10 complete episodes (Haiku reasoning, mid-chain: 1 of 10 started).

| fake verdicts | Sonnet | Haiku (reasoning) | Haiku (fast) |
|---|---|---|---|
| one, position 1 | 0.45 (0.45), $n = 20$ | 0.49 (0.43), $n = 30$ | 0.41 (0.43), $n = 30$ |
| two, positions 1–2 | 0.11 (0.00), $n = 30$ | 0.27 (0.00), $n = 30$ | 0.19 (0.00), $n = 30$ |
| three, positions 1–3 | 0.06 (0.00), $n = 20$ | 0.30 (0.00), $n = 29$ | 0.23 (0.00), $n = 30$ |
| two, positions 5–6 | 0.51 (0.54), $n = 16$ | – | 0.62 (0.59), $n = 30$ |

**Table 5: Frontier agents side with Bayes wherever it departs from counting, but Qwen3-14B and fast Haiku fail on peers' verdicts.** F1–F3: share of discriminating items (32 per family) on which the decision matches Bayes and not the count, and the largest mean $|P - P_B|$ across the three families. F4: share of cascade-verdict items (48) on which the stated $P$ is closer to Bayes than to the naive count, and the mean errors against each. There were 0 refusals. Llama needed 24 parse retries and left $P$ unparsed on 2 items.

| | decision = Bayes ≠ count | | | max $\lvert P - P_B \rvert$ | F4 cascade | |
|---|---|---|---|---|---|---|
| model | F1 prior | F2 prec. | F3 echo | F1–F3 | closer | err B / count |
| Opus 5.5 | 1.00 | 1.00 | 1.00 | 0.002 | 1.00 | 0.000 / 0.284 |
| GPT-6.1 | 1.00 | 1.00 | 1.00 | 0.000 | 1.00 | 0.002 / 0.286 |
| Gemini 4 Argon | 1.00 | 1.00 | 1.00 | 0.000 | 1.00 | 0.000 / 0.284 |
| Sonnet 5.5 | 1.00 | 1.00 | 1.00 | 0.002 | 0.98 | 0.015 / 0.286 |
| Haiku 5.5 (reasoning) | 1.00 | 0.97 | 0.97 | 0.002 | 0.92 | 0.081 / 0.307 |
| Haiku 5.5 (fast) | 1.00 | 1.00 | 1.00 | 0.005 | 0.44 | 0.211 / 0.208 |
| Qwen3-14B (reasoning) | 0.84 | 0.78 | 0.69 | 0.005 | 0.02 | 0.281 / 0.014 |
| Qwen3-14B (no-think) | 0.53 | 0.50 | 0.44 | 0.281 | 0.58 | 0.337 / 0.418 |
| Llama-3.1-8B | 0.44 | 0.62 | 0.47 | 0.401 | 0.68 | 0.354 / 0.455 |